

⇒ Web Page Layout and Linking

The social media application is designed with a structured layout to provide smooth navigation. When a user first visits the site, they are directed to the onboarding screen. This page contains two clear options: login for existing users and sign up for new users. After successful authentication, the user is redirected to the dashboard, which serves as the main hub of the application.

The dashboard follows a three-panel design. On the left is a fixed sidebar that contains links to the home page (feed), create post page, conversations and the profile page. The profile section also includes the options to switch accounts or log out, consolidating all account-related actions. The center panel is dynamic and changes based on navigation: the home page shows a feed of posts from followed users, each allowing comments, reactions and at most one multimedia attachment; the create post page provides a text box and media upload option; the conversations page lists chats and loads threaded discussions with multiple participants; and the profile page shows the user's details and their post history. When another user is searched, the main panel displays that user's posts in chronological order.

The right-hand panel is dedicated to user discovery. A search bar at the top displays matching users as text is typed. Selecting a user updates the right panel with their profile details and follow options, while the main panel shows their post history. Below this, a "People You May Know" section provides suggestions. This section remains visible at the bottom of the right panel and refreshes dynamically, showing general recommendations if no search is active, or related profiles if a specific user has been searched.

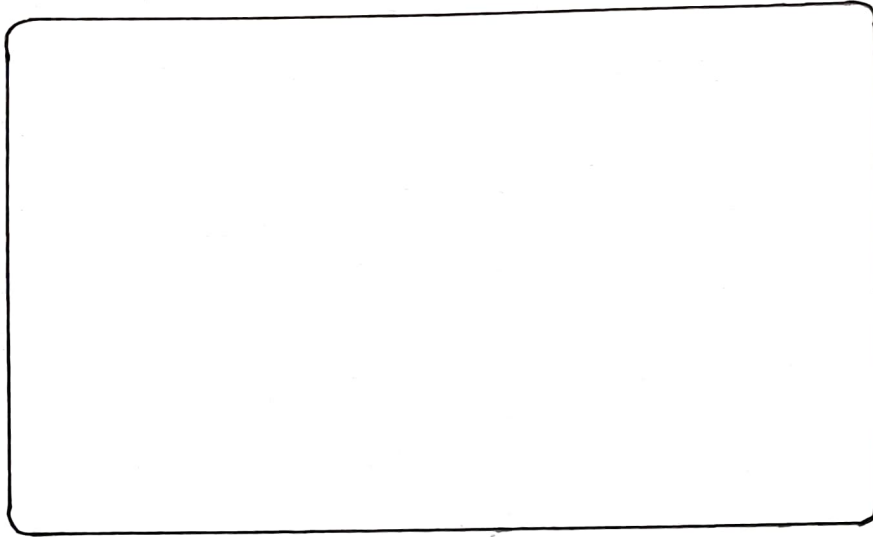
Navigation between pages is seamless. The sidebar enables quick switching between home, post creation, conversations and the profile. Clicking on usernames or avatars in posts and conversations directs the user to the relevant profile. The right panel anchors the interactive experience by handling search, follow requests, and contextual suggestions, while the main panel adapts to display the corresponding content.

Together, the onboarding screen, sidebar, dynamic main panel, and interactive right panel provide a consistent and cohesive user experience, allowing smooth movement between authentication, content viewing, posting, conversations and social discovery.

LOGIN

[G](#) [X](#) [Sign Up](#)

[SIGN IN](#)



Q Search

People you may
know.....

 User 1

 User 2

 User 3

⋮



☐ Conversation 1

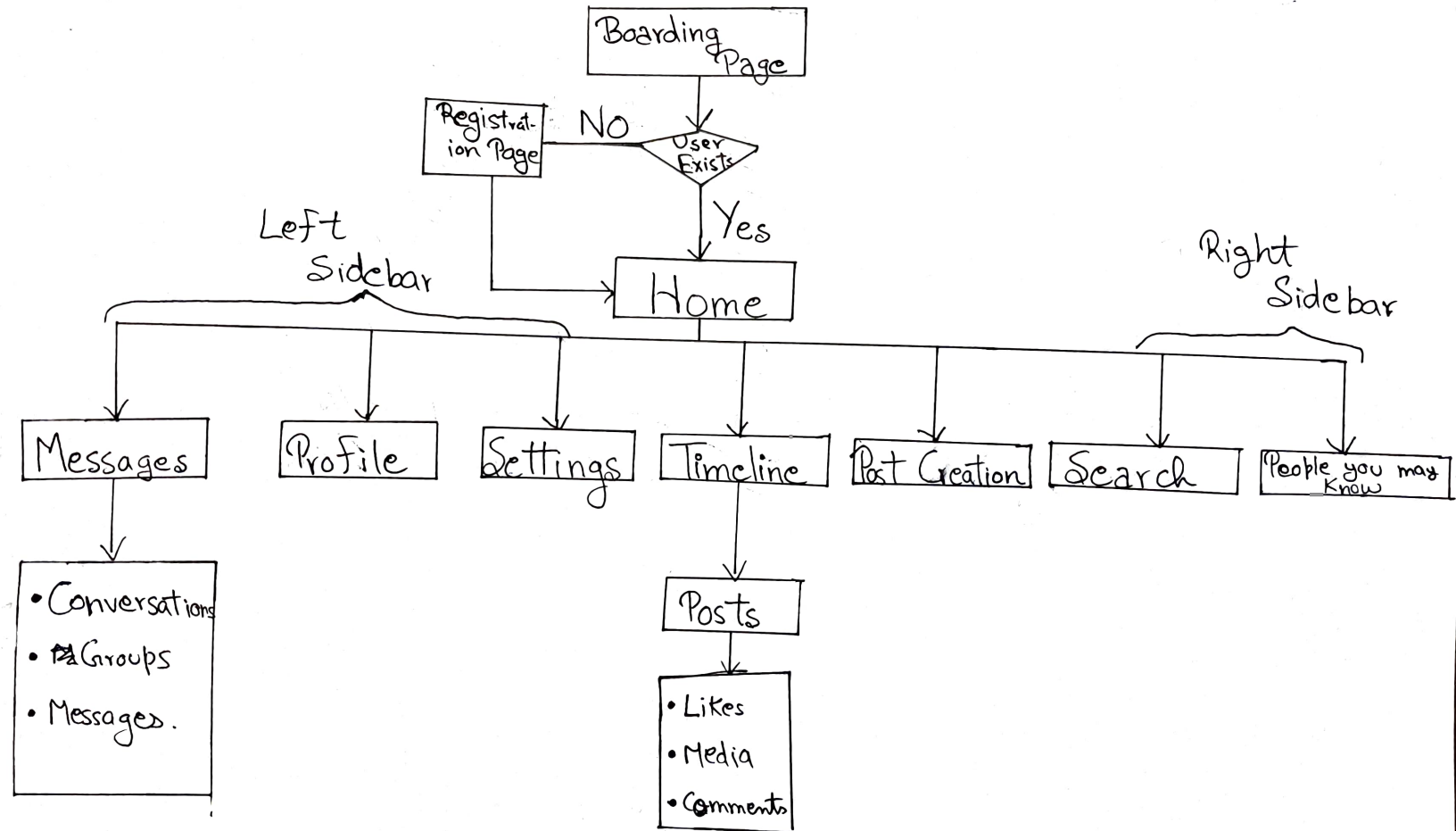
☐ Conversation 2

☐ Conversation 3

Today



1hr ago



⇒ Database Tables

User

- user_id, primary key, ~~not~~ int
- username, varchar
- email_id, varchar
- first_name, varchar
- last_name, varchar
- ^{pic}media_url, varchar
- bio, varchar
- dob, date
- mobile, varchar
- alternate_mail, varchar
- privacy_settings, jsonb
- password (hashed), varchar
- country, varchar
- state, varchar
- city, varchar

Post

- post_id, primary key, int
- user_id, foreign key, int
- content, text
- media_url, varchar
- timestamp, datetime
- visibility, enum("public", "private") default "public"
- latitude, decimal
- longitude, decimal

Comment

- comment_id, primary key, int
- user_id, foreign key, int
- post_id, foreign key, int
- content, text
- timestamp, datetime
- parent_comment_id, foreign key, int

Reaction

- post_id, foreign key, int
 - user_id, foreign key, int
 - reaction_id, foreign key, int
- } primary key

Reaction-Lookup

- reaction_id, primary key, int
- reaction, text (stored as SVG)

~~Message~~ Conversation

- ~~message~~ conversation_id, primary key, int
- participants, json
- conversation_settings, json
- timestamp, datetime

Message

- conversation_id, foreign key, int
- sender_id, foreign key (user_id), int
- message, text
- timestamp, datetime

primary key: {conversation_id, sender_id, timestamp}

Following

- following_id, foreign key (user_id), int
- follower_id, foreign key (user_id), int
- status, enum ("following", "follow", "pending") default "follow"
- timestamp, datetime

Friendship

- user_id1, foreign key(user_id), int
- user_id2, foreign key(user_id), int
- num_messages, int
- num_reactions, int
- num_comments, int
- friend_score, int

⇒ Entity-Relationship (ER) Diagram

